

FSHS224

## Sodium Benzoate (NF/FCC)

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 苯甲酸钠  
**Sodium Benzoate (NF/FCC)**

**Cat No. :** S224-500  
**Synonyms** Benzoic acid, sodium salt; Sodium benzoate.  
**CAS No** 532-32-1  
**Molecular Formula** C7H5O2Na

**Supplier** Fisher Scientific Company  
 One Reagent Lane  
 Fair Lawn, NJ 07410  
 Tel: (201) 796-7100

**Emergency Telephone Number** CHEMTREC®, Inside the USA: 800-424-9300  
 CHEMTREC®, Outside the USA: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Powder Solid

**Appearance**  
White

**Odor**  
Odorless

#### Emergency Overview

May be harmful if swallowed. Causes serious eye irritation. Hygroscopic.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 5 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements



**Signal Word**

**Warning**

#### Hazard Statements

H303 - May be harmful if swallowed  
 H319 - Causes serious eye irritation

# SAFETY DATA SHEET

## Sodium Benzoate (NF/FCC)

### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear eye protection/ face protection

#### Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Physical and Chemical Hazards

Hygroscopic.

#### Health Hazards

May be harmful if swallowed. Causes serious eye irritation.

#### Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| Sodium benzoate | 532-32-1 | 100      |

## SECTION 4. FIRST AID MEASURES

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Ingestion

Get medical attention if symptoms occur. Clean mouth with water and drink afterwards plenty of water.

#### Most important symptoms and effects

None reasonably foreseeable.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

#### Notes to Physician

Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

# SAFETY DATA SHEET

## Sodium Benzoate (NF/FCC)

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

### SECTION 7. HANDLING AND STORAGE

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Protect from moisture.

**Specific Use(s)**

Use in laboratories

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters**

| Component       | ACGIH TLV                          | OSHA PEL | NIOSH | The United Kingdom | European Union |
|-----------------|------------------------------------|----------|-------|--------------------|----------------|
| Sodium benzoate | TWA: 2.5 mg/m <sup>3</sup><br>Skin |          |       | -                  |                |

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the

## Sodium Benzoate (NF/FCC)

workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                   |                          |                                          |
|-----------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                 | White                    |                                          |
| <b>Physical State</b>             | Powder Solid             |                                          |
| <b>Odor</b>                       | Odorless                 |                                          |
| <b>Odor Threshold</b>             | No data available        |                                          |
| <b>pH</b>                         | ~ 8                      |                                          |
| <b>Melting Point/Range</b>        | 436 °C / 816.8 °F        |                                          |
| <b>Softening Point</b>            | No data available        |                                          |
| <b>Boiling Point/Range</b>        | 465 °C / 869 °F          |                                          |
| <b>Flash Point</b>                | Not applicable           | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>           | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>   | No information available |                                          |
| <b>Explosion Limits</b>           | No data available        |                                          |
| <b>Vapor Pressure</b>             | negligible               |                                          |
| <b>Vapor Density</b>              | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b> | 1.44 - 1.5 @ 20°C        |                                          |

## SAFETY DATA SHEET

## Sodium Benzoate (NF/FCC)

|                                                |                          |
|------------------------------------------------|--------------------------|
| <b>Bulk Density</b>                            | No data available        |
| <b>Water Solubility</b>                        | Soluble                  |
| <b>Solubility in other solvents</b>            | No information available |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |
| <b>Component</b>                               | <b>log Pow</b>           |
| Sodium benzoate                                | -2.13                    |
| <b>Autoignition Temperature</b>                | 540 °C / 1004 °F         |
| <b>Decomposition Temperature</b>               | 450-475 °C               |
| <b>Viscosity</b>                               | Not applicable           |
| <b>Explosive Properties</b>                    | No information available |
| <b>Oxidizing Properties</b>                    | No information available |

|                          |          |
|--------------------------|----------|
| <b>Molecular Formula</b> | C7H5O2Na |
| <b>Molecular Weight</b>  | 144.11   |
| <b>VOC Content(%)</b>    | 100      |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions. Hygroscopic.                                                                    |
| <b>Hazardous Reactions</b>      | None under normal processing.                                                                                   |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                                                        |
| <b>Conditions to Avoid</b>      | Avoid dust formation. Incompatible products. Excess heat. Exposure to moisture. Exposure to moist air or water. |
| <b>Materials to avoid</b>       | Strong oxidizing agents. Acids.                                                                                 |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component       | LD50 Oral                 | LD50 Dermal            | LC50 Inhalation        |
|-----------------|---------------------------|------------------------|------------------------|
| Sodium benzoate | LD50 = 4070 mg/kg ( Rat ) | LD50 >2000 mg/kg (Rat) | LC50 = 12.2 mg/l (Rat) |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

| Component                           | Test method                                       | Test species | Study result    |
|-------------------------------------|---------------------------------------------------|--------------|-----------------|
| Sodium benzoate<br>532-32-1 ( 100 ) | OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                           | Test method                                             | Test species         | Study result |
|-------------------------------------|---------------------------------------------------------|----------------------|--------------|
| Sodium benzoate<br>532-32-1 ( 100 ) | OECD Test Guideline 473<br>Chromosomal aberration assay | in vivo<br>Mammalian | negative     |

## Sodium Benzoate (NF/FCC)

|                                            |                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------|
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;                 | No data available                                                              |
| (h) STOT-single exposure;                  | No data available                                                              |
| (i) STOT-repeated exposure;                | No data available                                                              |
| Target Organs                              | No information available.                                                      |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                        |
| Symptoms / effects, both acute and delayed | No information available                                                       |

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

| Component       | Freshwater Fish                                                                                                    | Water Flea                            | Freshwater Algae                                        | Microtox             |
|-----------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------|----------------------|
| Sodium benzoate | LC50: 420 - 558 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 100 mg/L, 96h static (Pimephales promelas) | EC50: < 650 mg/L, 48h (Daphnia magna) | EC50: > 30.5mg/l, 72h (Pseudokirchneriella subcapitata) | EC50 = 500 mg/L 24 h |

## Persistence and Degradability

## Persistence

Persistence is unlikely.

| Component                           | Degradability |
|-------------------------------------|---------------|
| Sodium benzoate<br>532-32-1 ( 100 ) | 94% (28d)     |

## Bioaccumulative Potential

Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Sodium benzoate | -2.13   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

## Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

## Sodium Benzoate (NF/FCC)

|                               |                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                     |
| <b>Other Information</b>      | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. |

## SECTION 14. TRANSPORT INFORMATION

**Road and Rail Transport** Not Regulated

**IMDG/IMO** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component       | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Sodium benzoate | -                                                   | -                                       | X    | X     | 208-534-8 | X    | X   | X     | X    | X    | X    | KE-02711 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 15-May-2014  
**Revision Date** 15-May-2024  
**Revision Summary** Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**PNEC** - Predicted No Effect Concentration  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**